ARYA SREE ASHOK

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EDUCATION

Paavai Engineering College

Bachelor of Engineering (B.E.) in Computer Science – AI & ML | CGPA: 8.36
Expected Graduation: 2027

TECHNICAL SKILLS

- **Programming Languages**: Python, Java, C++
- **AI/ML Frameworks & Tools**: OpenCV, MediaPipe, NumPy, PyTorch/TensorFlow, Stable Diffusion, IDM-VTON, CUDA
- **Core Research Areas**: Computer Vision (CV), Deep Learning (DL), Natural Language Processing (NLP), Generative AI, Image Processing
- **Languages:** English, Malayalam, Tamil

TECHNICAL & RESEARCH PROJECTS

Tamil Sign Language to Speech Converter

- Research Focus: Computer Vision, Deep Learning, and Real-Time Accessibility Systems
- Engineered a real-time Tamil Sign Language recognition pipeline using MediaPipe for hand landmark extraction and OpenCV for spatial processing.
- Conducted comparative analysis and performance evaluation across multiple architectures, including VGG, ResNet, ConvNeXt, and MLP, for gesture classification.
- Optimized data preprocessing techniques to achieve an accuracy of 80-90% utilizing a fine-tuned MLP model.

AI-Powered Wardrobe Styling System

- Research Focus: Generative AI, Multimodal Embeddings, and Clustering
- Engineered an outfit recommendation engine leveraging CLIP multimodal embeddings for semantic matching and K-means clustering for color harmony analysis.
- Implemented a virtual try-on conditional image generation pipeline utilizing Stable Diffusion v1.5 integrated with IDM-VTON to render realistic garment transformations.
- Collaborative research and development conducted with Claude AI to architect a modular, GPU-accelerated framework (RTX 5060, CUDA) optimizing real-time feature extraction, pose estimation, and outfit ranking.

Smart Intrusion Detection and Alert System

- Research Focus: Edge Computing & Intelligent Surveillance
- Designed an automated, real-time intrusion monitoring pipeline utilizing computer vision techniques to detect unauthorized access.
- Integrated automated trigger-response modules to capture image payloads upon detection events and dynamically transmit encrypted email alerts

EXPERIENCE

Research Intern – Computer Vision Techniques

NSIC – Chennai | June 2025

- Researched facial recognition methodologies using Python, OpenCV, NumPy, and imutils to optimize real-time detection accuracy.
- Explored and applied image preprocessing techniques to enhance model robustness against lighting variations and facial orientations in live environments.
- Contributed to profiling and streamlining inference workflows to improve computational efficiency.

Poster Designer & Instagram Manager

Arttifai Tech | March 2024 – August 2024

- Designed high-impact posters and visual assets using creative design tools to elevate brand presence.
- Managed end-to-end Instagram channels, driving consistent audience growth through daily content scheduling and page curation.

CERTIFICATIONS & WORKSHOPS

- **AWS Certified AI Practitioner Challenge** – Successfully participated in/completed the AWS AI Practitioner challenge, building foundational knowledge in cloud-based AI/ML workloads, generative AI concepts, and AWS machine learning services.
- **Google Prompting Essentials Specialization (Coursera)** – Specialized in advanced prompting frameworks, data analysis automation, and workflow optimization.
- **Google AI Essentials Specialization (Coursera)** – Focused on scaling productivity via cross-functional AI integration and maintaining responsible AI principles.
- **Generative AI Workshop (PSG College)** – Immersive training on Generative Adversarial Networks (GANs), Large Language Models (LLMs), and interactive prompt testing with ChatGPT/Gemini.
